

Chapter 12

Maximum and Minimum Values

§ Section: 4.1

One of the most significant applications of derivatives lies in solving optimization problems, where the objective is to determine the maximum or minimum values of a function. To approach these problems, we first define the notions of extreme values and then examine how derivatives can be employed to identify them.

12.1 Definition

Definition:

Let c be a number in the domain D of a function f . Then $f(c)$ is the

1. **absolute maximum** value of f on D if $f(c) \geq f(x)$ for x in D .
2. **absolute minimum** value of f on D if $f(c) \leq f(x)$ for x in D .

Definition:

The number $f(c)$ is a

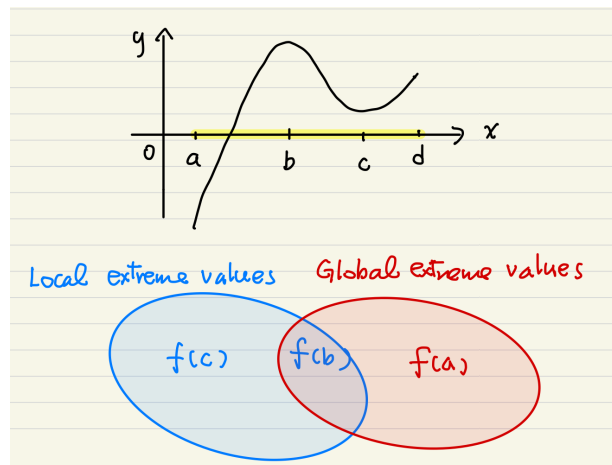
1. **local maximum** value of f if $f(c) \geq f(x)$ when x is near c .
2. **local minimum** value of f if $f(c) \leq f(x)$ when x is near c .

Remark:

1. An absolute maximum or minimum is sometimes called a **global** maximum or minimum.
2. The maximum and minimum values of f are called **extreme values** of f .
3. In above Definition, if we say that something is true **near** c , we mean that it is true on some open interval containing c . Hence, the local extreme values always occur in the **interior** of the domain of f .

Example 1.1

The graph of $f(x)$ is given as below. Find the **local extreme values** and **global extreme values** of $f(x)$.

**The Extreme Value Theorem**

If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers c and d in $[a, b]$.

Remark:

Be careful about the requirements of the Extreme Value Theorem: f is **continuous** on a **closed and bounded** interval. If requirements are not satisfied, the conclusion may not be true.

- $f(x)$ is continuous: Ex: $f(x) = \begin{cases} \tan x, & x \neq \frac{\pi}{2} \\ 0, & x = \frac{\pi}{2} \end{cases}$ on $[0, \pi]$, $f(x) = \begin{cases} x, & 0 \leq x < \frac{1}{2} \\ 0, & x = \frac{1}{2} \\ x-1, & \frac{1}{2} < x \leq 1 \end{cases}$ on $[0, 1]$.
- The interval is closed and bounded: Ex: $f(x) = x^3$ on $\mathbb{R}$, Ex: $f(x) = x$ on $(0, 1)$.

Fermat's Theorem

If f has a local maximum or minimum at c , and if $f'(c)$ exists, then $f'(c) = 0$.

Warning: Common mistakes of applying Fermat's Theorem

- If $f(x)$ has local extreme value at c , then $f'(c) = 0$. $f(x) = |x|$ at $x=0$
 f is NOT diff at $x=0$

- If $f'(c) = 0$, then $f(x)$ attains local extreme value at c . $f(x) = x^3$ at $x=0$.
 $f'(0) = 0$ but $f(0)$ is NOT a local extreme value.

Example 1.2

Prove Fermat's Theorem.

pf: Assume that $f(c) \geq f(x)$ for $x \in (c-\delta, c+\delta)$.

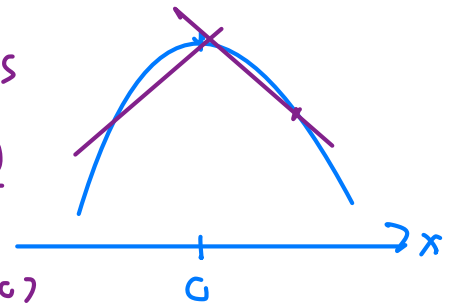
$$\lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c} \leq 0$$

$$\lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c} \geq 0$$

$\therefore f'(c)$ exists

$$\therefore \lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c} = \lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c} = 0$$

$$= \lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c} = 0$$



Fermat's Theorem does suggest that we should at least start looking for extreme values of f at the numbers c where $f'(c) = 0$ or where $f'(c)$ does not exist. Such numbers are given a special name.

Definition:

A **critical number** of a function f is a number c in the domain of f such that either $f'(c) = 0$ or $f'(c)$ does not exist.

Theorem:

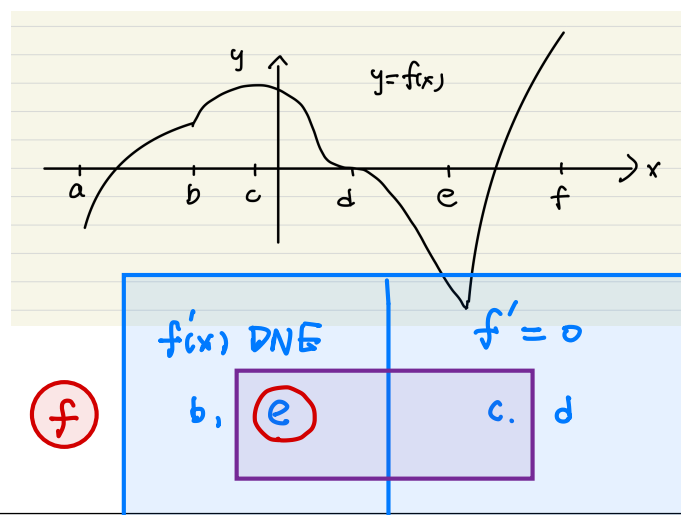
If f has a local maximum or minimum at c , then c is a critical number of f .

Example 1.3

The graph of $f(x)$ is given as below, Find critical numbers of $f(x)$ and local/global extreme values.

local
extreme
value

global
extreme
value



Exercise 1.1

The profit function is defined as $\Pi(x) = R(x) - C(x)$, where $R(x)$ is the total revenue function, $C(x)$ is the total cost function, and x is the number of units of a product produced and sold. Assume that $R(x)$ and $C(x)$ are differentiable. Show that at the level of production x_0 that yields the maximum profit the following condition is satisfied:

$$R'(x_0) = C'(x_0).$$

Summary

To find an absolute maximum or minimum of a **continuous** function on a **closed** interval, we note that either it is local or it occurs at an endpoint of the interval.

The Closed Interval Method

To find the *absolute* maximum and minimum values of a continuous function f on a closed interval $[a, b]$:

1. Find the values of f at the critical numbers of f in (a, b) .
2. Find the values of f at the endpoints of the interval.
3. The largest of the values from Steps 1 and 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

Example 1.4

Find absolute extreme values of $f(x) = x^2 - 3|x - 1|$ on $[-2, 3]$.

Sol: **Step 1** f is NOT diff at $x=1$.

$$f(1) = 1$$

For $x > 1$, $f(x) = x^2 - 3(x-1)$

$$f\left(\frac{3}{2}\right) = \frac{9}{4} - 3 \cdot \frac{1}{2} = \frac{3}{4}$$

Solve

$$f'(x) = 2x - 3 = 0 \\ \Rightarrow x = \frac{3}{2}$$

For $x < 1$, $f(x) = x^2 - 3(1-x)$. Solve $f'(x) = 2x + 3 = 0 \\ \Rightarrow x = -\frac{3}{2}$.

$$f\left(-\frac{3}{2}\right) = \frac{9}{4} - 3 \cdot \frac{5}{2} = -\frac{21}{4}$$

Step 2

$$f(-2) = 4 - 9 = -5$$

$$f(3) = 9 - 6 = 3$$

Step 3

Global Max: $f(3) = 3$

$$\therefore \text{Min?} = \frac{85}{85} f\left(-\frac{3}{2}\right) = -5\frac{1}{4}$$

Example 1.5Find absolute extreme values of $f(x) = x^{\frac{1}{3}}(x-2)$ on $[-1, 2]$.

Sol: **Step 1** $f'(x) = \frac{1}{3}x^{-\frac{2}{3}}(x-2) + x^{\frac{1}{3}} = \frac{(x-2) + 3x}{3x^{2/3}} = \frac{4x-2}{3x^{2/3}}$

$$f'(x) = 0 \Rightarrow x = \frac{1}{2} . \quad f\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^{\frac{1}{3}} \cdot \left(-\frac{3}{2}\right) = -\frac{3}{2\sqrt[3]{2}}$$

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{x-2}{x^{2/3}} = -\infty \Rightarrow f'(0) \text{ DNE !} \quad f(0) = 0$$

Step 2

$f(-1) = 3$

$f(2) = 0$

Step 3

Global Max: $f(-1) = 3$

Min: $f\left(\frac{1}{2}\right) = -\frac{3}{2\sqrt[3]{2}}$

Example 1.6Find absolute extreme values of $f(x) = 2\sin x - \sin 2x$ on the interval $[-\frac{\pi}{2}, \pi]$.

Sol: **Step 1** Solve $f'(x) = 2\cos x - \cos 2x \cdot 2 = 0$

$$\Rightarrow 2\cos x - 2(2\cos^2 x - 1) = 0$$

$$\Rightarrow \cos x - 2\cos^2 x + 1 = 0$$

$$\Rightarrow 2\cos^2 x - \cos x - 1 = 0$$

$$\Rightarrow (2\cos x + 1)(\cos x - 1) = 0 \Rightarrow \cos x = -\frac{1}{2} \text{ or } 1.$$

$$\Rightarrow x = \frac{2}{3}\pi, 0, \quad \begin{aligned} f\left(\frac{2}{3}\pi\right) &= \sqrt{3} + \frac{\sqrt{3}}{2} = \frac{3}{2}\sqrt{3} \\ f(0) &= 0 \end{aligned}$$

Step 2

$f\left(-\frac{\pi}{2}\right) = -2, \quad f(\pi) = 0$

Step 3

Global max: $f\left(\frac{2}{3}\pi\right) = \frac{3}{2}\sqrt{3}$

Min: $f\left(-\frac{\pi}{2}\right) = -2.$